

Bayesian Z-residual Diagnostics for Dolphin-Sighting Counts (demo: section 4)

A real-data discrete regression example

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1 Overview

This demo illustrates a practical workflow for Bayesian model diagnostics with Z-residuals in a real discrete-response regression problem. The response variable is a count with many zeros, so we compare Poisson, negative binomial, hurdle Poisson, and hurdle negative binomial models. The workflow first checks the overall residual distribution, then examines residual patterns against covariates, especially water depth. We also compare Z-residual diagnostics with posterior predictive checks (PPC), holdout predictive checks (HPC), and LOO expected log predictive density.

2 Installing Zresidual and Other packages

2.1 Installing Zresidual from the Source

```
# Ensures the path starts at the project root regardless of the current working directory
renv::remove("Zresidual")
renv::install(here::here(), dependencies = TRUE)
```

2.2 Installing and Loading R Packages Used in This Demo

```

# Vector of required packages
pkgs <- c(
  "brms", "statip", "VGAM", "plotrix", "actuar",
  "stringr", "Rlab", "dplyr", "rlang", "tidyr",
  "matrixStats", "timeDate", "katex", "gt", "loo"
)

# Check for missing packages and install if needed
missing_pkgs <- pkgs[!pkgs %in% installed.packages()[, "Package"]]
if (length(missing_pkgs) > 0) {
  message("Installing missing packages: ", paste(missing_pkgs, collapse = ", "))
  renv::install(missing_pkgs, dependencies = TRUE)
} else {
  message("All required packages are already installed.")
}

# Load all packages
invisible(lapply(pkgs, library, character.only = TRUE))

options(mc.cores = parallel::detectCores())

library(Zresidual)

```

3 Load the Dolphin-Sighting Dataset

This demo analyzes a dolphin-sighting count dataset with $n = 936$ observations. The response is the count of dolphin sightings. The covariates include water depth (`depth`), water temperature (`temp`), distance to the river mouth (`dr`, indexing the estuarine-coastal gradient), `salinity`, `turbidity`, survey `year`, and a location identifier.

```

data_path <- file.path(resources_path, "DolphinsThailandV2.csv")

if (!file.exists(data_path)) {
  stop(
    "Cannot find the dolphin-sighting dataset: ", data_path,
    "
Expected file name: DolphinsThailandV2.csv",
    call. = FALSE
  )
}

```

```

)
}

Dolphins <- read.csv(data_path)
Dolphins$year <- as.factor(Dolphins$year)
prop_zero <- mean(Dolphins$sightings == 0, na.rm = TRUE)

# proportion of zeros
cat("Proportion of Zeros:", prop_zero)

```

Proportion of Zeros: 0.7564103

4 Diagnosing Four Linear Models

4.1 Fitting Hurdle NB, Hurdle Poisson, NB, and Poisson Models

Because zeros are prevalent, we fit four linear count models: hurdle Poisson (HP), hurdle negative binomial (HNB), Poisson, and negative binomial (NB). These models provide a simple starting point for checking both the response family and the assumed linear covariate effects.

```

# The model fitting takes a long time. Hence the fitted model object is provided.
filename <- file.path(resources_path, "fit_Dolphins.Rdata")

if (forcerefit || !file.exists(filename)) {
  Dolphins_count_formula <- formula(paste("sightings ~ depth + temp + dr + salinity + turbidity +
  Dolphins_zero_formula <- formula(paste("hu ~ depth + temp + dr + salinity + turbidity + year ")
  Dolphins_hnb <- brm(
    bf(Dolphins_count_formula, Dolphins_zero_formula),
    data = Dolphins,
    family = hurdle_negbinomial(link = "log", link_shape = "identity", link_hu = "logit"),
    save_pars = save_pars(all = TRUE),
    refresh = 0
  )

  Dolphins_hp <- brm(
    bf(Dolphins_count_formula, Dolphins_zero_formula),
    data = Dolphins,
    family = hurdle_poisson(link = "log", link_hu = "logit"),

```

```

    save_pars = save_pars(all = TRUE),
    refresh   = 0
  )

Dolphins_nb <- brm(
  Dolphins_count_formula,
  data      = Dolphins,
  family    = brms::negbinomial(link = "log", link_shape = "identity"),
  save_pars = save_pars(all = TRUE),
  refresh   = 0
)

Dolphins_pois <- brm(
  Dolphins_count_formula,
  data      = Dolphins,
  family    = poisson(link = "log"),
  save_pars = save_pars(all = TRUE),
  refresh   = 0
)

save(Dolphins_hnb, Dolphins_hp, Dolphins_nb, Dolphins_pois, file = filename)

} else {
  base::load(filename)
}

```

4.2 Computing Z-Residuals (Posterior)

We first fit each family with linear covariate effects and evaluate overall GOF using **posterior RSP Z-residuals**.

```

library(Zresidual)

zres_linear_post_file <- file.path(resources_path, "zres_linear_post_list_all.rds")
zres_linear_iscv_file <- file.path(resources_path, "zres_linear_iscv_list_all.rds")

required_linear_zres_names <- c("HNB", "HP", "NB", "Poisson")

if (

```

```

!force_recompute_zres &&
  file.exists(zres_linear_post_file) &&
  file.exists(zres_linear_iscv_file)
) {
  zres_linear_post_list <- readRDS(zres_linear_post_file)
  zres_linear_iscv_list <- readRDS(zres_linear_iscv_file)

  missing_post <- setdiff(required_linear_zres_names, names(zres_linear_post_list))
  missing_iscv <- setdiff(required_linear_zres_names, names(zres_linear_iscv_list))

  if (length(missing_post) || length(missing_iscv)) {
    stop(
      "Saved linear Z-residual files do not contain the expected model names.",
      "Missing from posterior list: ", paste(missing_post, collapse = ", "), "
      "Missing from ISCV list: ", paste(missing_iscv, collapse = ", "),
      call. = FALSE
    )
  }
} else {
  zres_linear_post_list <- list(
    HNB = Zresidual(Dolphins_hnb, data = Dolphins, mcmc_summarize = "post", nrep = nrep_z),
    HP = Zresidual(Dolphins_hp, data = Dolphins, mcmc_summarize = "post", nrep = nrep_z),
    NB = Zresidual(Dolphins_nb, data = Dolphins, mcmc_summarize = "post", nrep = nrep_z),
    Poisson = Zresidual(Dolphins_pois, data = Dolphins, mcmc_summarize = "post", nrep = nrep_z)
  )

  zres_linear_iscv_list <- list(
    HNB = Zresidual(Dolphins_hnb, data = Dolphins, mcmc_summarize = "iscv", nrep = nrep_z),
    HP = Zresidual(Dolphins_hp, data = Dolphins, mcmc_summarize = "iscv", nrep = nrep_z),
    NB = Zresidual(Dolphins_nb, data = Dolphins, mcmc_summarize = "iscv", nrep = nrep_z),
    Poisson = Zresidual(Dolphins_pois, data = Dolphins, mcmc_summarize = "iscv", nrep = nrep_z)
  )

  saveRDS(zres_linear_post_list, zres_linear_post_file)
  saveRDS(zres_linear_iscv_list, zres_linear_iscv_file)
}

```

```

Dolphins_zres_post_rpp_hnb <- zres_linear_post_list$HNB
Dolphins_zres_post_rpp_hp  <- zres_linear_post_list$HP
Dolphins_zres_post_rpp_nb  <- zres_linear_post_list$NB
Dolphins_zres_post_rpp_pois <- zres_linear_post_list$Poisson

Dolphins_zres_iscv_rpp_hnb <- zres_linear_iscv_list$HNB
Dolphins_zres_iscv_rpp_hp  <- zres_linear_iscv_list$HP
Dolphins_zres_iscv_rpp_nb  <- zres_linear_iscv_list$NB
Dolphins_zres_iscv_rpp_pois <- zres_linear_iscv_list$Poisson

```

4.3 Inspecting the Normality of Z-Residuals for Checking Overall GOF

Figure 1 shows QQ-plots of posterior RSP Z-residuals. The QQ-plots of the HNB and NB models adhere closely to the diagonal lines, suggesting adequate fit, whereas the QQ-plots of the HP and Poisson models deviate markedly from the diagonal lines, especially in their upper tails. The $Z^{\text{Post-RSP}}\text{-SW}$ p -values beside the QQ-plots also confirm numerically these impressions: HNB and NB models having p -values ≥ 0.05 (no rejection) versus HP and Poisson both having < 0.01 p -values (clear rejection). It is clear that the Poisson family cannot model well the strong overdispersion and the hurdle modification does not improve it.

```

par(mfrow = c(2,2))
qqnorm(Dolphins_zres_post_rpp_hnb,irep = 2,
       main=expression("(a) HNB Linear, " * Z^"Post-RSP" * " QQ Plot"),
       cex.main=1.5,cex.lab=1.5, cex.axis=1.5, my.mar = c(5, 4, 4, 6) + 0.1,cex = 1.0)
qqnorm(Dolphins_zres_post_rpp_nb,irep = 2,
       main=expression("(b) NB Linear, " * Z^"Post-RSP" * " QQ Plot"),
       cex.main=1.5,cex.lab=1.5, cex.axis=1.5, my.mar = c(5, 4, 4, 6) + 0.1,cex = 1.0)
qqnorm(Dolphins_zres_post_rpp_hp,irep = 2,
       main=expression("(c) HP Linear, " * Z^"Post-RSP" * " QQ Plot"),
       cex.main=1.5,cex.lab=1.5, cex.axis=1.5, my.mar = c(5, 4, 4, 6) + 0.1,cex = 1.0)
qqnorm(Dolphins_zres_post_rpp_pois,irep = 2,
       main=expression("(d) Poisson Linear, " * Z^"Post-RSP" * " QQ Plot"),
       cex.main=1.5,cex.lab=1.5, cex.axis=1.5, my.mar = c(5, 4, 4,6) + 0.1,cex = 1.0)
par(mfrow = c(1,1))

```

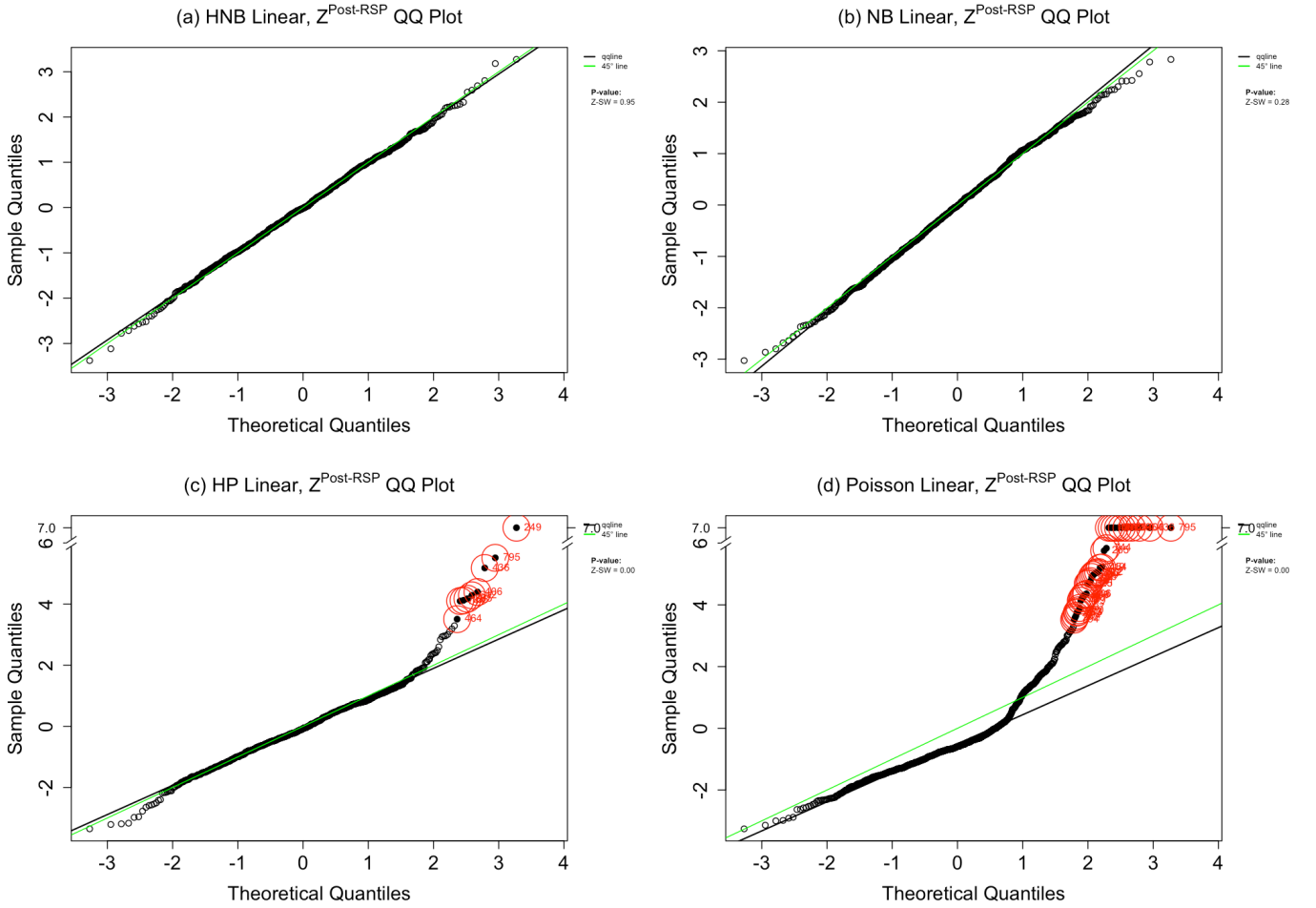


Figure 1: QQ plots of posterior RSP Z-residuals for the (a) HNB, (b) NB, (c) HP, and (d) Poisson models, each fitted in linear form to the dolphin-sighting dataset.

4.4 Diagnostic Tests

For each candidate model (HNB, NB, HP, Poisson) we compute posterior Z-residuals and conduct: (i) marginal normality via the Shapiro–Wilk test; (ii) functional-form checks via ANOVA of Z-residuals across strata of the covariate `depth`; and (iii) an omnibus statistic $\chi_z^2 = \sum_{i=1}^n z_i^2$ compared to χ_ν^2 with $\nu = n - p_{\text{eff}}$, where p_{eff} is the effective number of parameters used in WAIC and LOOCV approximations. For discrete outcomes, we re-randomize the auxiliary uniforms $J=500$ times, compute the replicated test p -values, and report a single conservative summary via the distribution-free upper-bound p -value (`upper_bound_pvalue`).

```
pv_hnb <- sw.test.zresid(Dolphins_zres_post_rpp_hnb)
pv_nb  <- sw.test.zresid(Dolphins_zres_post_rpp_nb)
pv_hp  <- sw.test.zresid(Dolphins_zres_post_rpp_hp)
pv_pois <- sw.test.zresid(Dolphins_zres_post_rpp_pois)
```

```

depth_aov_hnb <- aov.test.zresid(Dolphins_zres_post_rpp_hnb, X = "depth")
depth_aov_nb <- aov.test.zresid(Dolphins_zres_post_rpp_nb, X = "depth")
depth_aov_hp <- aov.test.zresid(Dolphins_zres_post_rpp_hp, X = "depth")
depth_aov_pois <- aov.test.zresid(Dolphins_zres_post_rpp_pois, X = "depth")

lp_aov_hnb <- aov.test.zresid(Dolphins_zres_post_rpp_hnb, X = "lp")
lp_aov_nb <- aov.test.zresid(Dolphins_zres_post_rpp_nb, X = "lp")
lp_aov_hp <- aov.test.zresid(Dolphins_zres_post_rpp_hp, X = "lp")
lp_aov_pois <- aov.test.zresid(Dolphins_zres_post_rpp_pois, X = "lp")

pv_hnb_iscv <- sw.test.zresid(Dolphins_zres_iscv_rpp_hnb)
pv_nb_iscv <- sw.test.zresid(Dolphins_zres_iscv_rpp_nb)
pv_hp_iscv <- sw.test.zresid(Dolphins_zres_iscv_rpp_hp)
pv_pois_iscv <- sw.test.zresid(Dolphins_zres_iscv_rpp_pois)

depth_aov_hnb_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_hnb, X = "depth")
depth_aov_nb_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_nb, X = "depth")
depth_aov_hp_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_hp, X = "depth")
depth_aov_pois_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_pois, X = "depth")

lp_aov_hnb_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_hnb, X = "lp")
lp_aov_nb_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_nb, X = "lp")
lp_aov_hp_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_hp, X = "lp")
lp_aov_pois_iscv <- aov.test.zresid(Dolphins_zres_iscv_rpp_pois, X = "lp")

p_effective_hnb <- loo(Dolphins_hnb, moment_match = TRUE)$estimates["p_loo", "Estimate"]
p_effective_nb <- loo(Dolphins_nb, moment_match = TRUE)$estimates["p_loo", "Estimate"]
p_effective_hp <- loo(Dolphins_hp, moment_match = TRUE)$estimates["p_loo", "Estimate"]
p_effective_pois <- loo(Dolphins_pois, moment_match = TRUE)$estimates["p_loo", "Estimate"]

n_obs <- nrow(Dolphins_zres_post_rpp_hnb)
ncols <- min(
  500L,
  ncol(Dolphins_zres_post_rpp_hnb),
  ncol(Dolphins_zres_post_rpp_nb),
  ncol(Dolphins_zres_post_rpp_hp),
  ncol(Dolphins_zres_post_rpp_pois),
  ncol(Dolphins_zres_iscv_rpp_hnb),

```



```

ncol(Dolphins_zres_iscv_rpp_nb),
ncol(Dolphins_zres_iscv_rpp_hp),
ncol(Dolphins_zres_iscv_rpp_pois)
)

chisq_zres_post_rpp_hnb <- numeric(ncols)
chisq_zres_post_rpp_nb <- numeric(ncols)
chisq_zres_post_rpp_hp <- numeric(ncols)
chisq_zres_post_rpp_pois <- numeric(ncols)

chisq_zres_iscv_rpp_hnb <- numeric(ncols)
chisq_zres_iscv_rpp_nb <- numeric(ncols)
chisq_zres_iscv_rpp_hp <- numeric(ncols)
chisq_zres_iscv_rpp_pois <- numeric(ncols)

chisq_post_rpp_hnb <- numeric(ncols)
chisq_post_rpp_nb <- numeric(ncols)
chisq_post_rpp_hp <- numeric(ncols)
chisq_post_rpp_pois <- numeric(ncols)

chisq_iscv_rpp_hnb <- numeric(ncols)
chisq_iscv_rpp_nb <- numeric(ncols)
chisq_iscv_rpp_hp <- numeric(ncols)
chisq_iscv_rpp_pois <- numeric(ncols)

for (i in seq_len(ncols)) {
  chisq_zres_post_rpp_hnb[i] <- sum(Dolphins_zres_post_rpp_hnb[, i]^2, na.rm = TRUE)
  chisq_zres_post_rpp_nb[i] <- sum(Dolphins_zres_post_rpp_nb[, i]^2, na.rm = TRUE)
  chisq_zres_post_rpp_hp[i] <- sum(Dolphins_zres_post_rpp_hp[, i]^2, na.rm = TRUE)
  chisq_zres_post_rpp_pois[i] <- sum(Dolphins_zres_post_rpp_pois[, i]^2, na.rm = TRUE)

  chisq_zres_iscv_rpp_hnb[i] <- sum(Dolphins_zres_iscv_rpp_hnb[, i]^2, na.rm = TRUE)
  chisq_zres_iscv_rpp_nb[i] <- sum(Dolphins_zres_iscv_rpp_nb[, i]^2, na.rm = TRUE)
  chisq_zres_iscv_rpp_hp[i] <- sum(Dolphins_zres_iscv_rpp_hp[, i]^2, na.rm = TRUE)
  chisq_zres_iscv_rpp_pois[i] <- sum(Dolphins_zres_iscv_rpp_pois[, i]^2, na.rm = TRUE)

  chisq_post_rpp_hnb[i] <- pchisq(q = chisq_zres_post_rpp_hnb[i], df = n_obs - p_effective_hnb,
  chisq_post_rpp_nb[i] <- pchisq(q = chisq_zres_post_rpp_nb[i], df = n_obs - p_effective_nb,
  chisq_post_rpp_hp[i] <- pchisq(q = chisq_zres_post_rpp_hp[i], df = n_obs - p_effective_hp,

```

```

chisq_post_rpp_pois[i] <- pchisq(q = chisq_zres_post_rpp_pois[i], df = n_obs - p_effective_pois, lower.tail = FALSE)

chisq_iscv_rpp_hnb[i] <- pchisq(q = chisq_zres_iscv_rpp_hnb[i], df = n_obs, lower.tail = FALSE)
chisq_iscv_rpp_nb[i] <- pchisq(q = chisq_zres_iscv_rpp_nb[i], df = n_obs, lower.tail = FALSE)
chisq_iscv_rpp_hp[i] <- pchisq(q = chisq_zres_iscv_rpp_hp[i], df = n_obs, lower.tail = FALSE)
chisq_iscv_rpp_pois[i] <- pchisq(q = chisq_zres_iscv_rpp_pois[i], df = n_obs, lower.tail = FALSE)
}

pv_hnb = upper_bound_pvalue(pv_hnb)
pv_nb = upper_bound_pvalue(pv_nb)
pv_hp = upper_bound_pvalue(pv_hp)
pv_pois = upper_bound_pvalue(pv_pois)

depth_aov_hnb = upper_bound_pvalue(depth_aov_hnb)
depth_aov_nb = upper_bound_pvalue(depth_aov_nb)
depth_aov_hp = upper_bound_pvalue(depth_aov_hp)
depth_aov_pois = upper_bound_pvalue(depth_aov_pois)

lp_aov_hnb = upper_bound_pvalue(lp_aov_hnb)
lp_aov_nb = upper_bound_pvalue(lp_aov_nb)
lp_aov_hp = upper_bound_pvalue(lp_aov_hp)
lp_aov_pois = upper_bound_pvalue(lp_aov_pois)

chisq_hnb_cap = upper_bound_pvalue(chisq_post_rpp_hnb)
chisq_nb_cap = upper_bound_pvalue(chisq_post_rpp_nb)
chisq_hp_cap = upper_bound_pvalue(chisq_post_rpp_hp)
chisq_pois_cap = upper_bound_pvalue(chisq_post_rpp_pois)

pv_hnb_iscv = upper_bound_pvalue(pv_hnb_iscv)
pv_nb_iscv = upper_bound_pvalue(pv_nb_iscv)
pv_hp_iscv = upper_bound_pvalue(pv_hp_iscv)
pv_pois_iscv = upper_bound_pvalue(pv_pois_iscv)

depth_aov_hnb_iscv = upper_bound_pvalue(depth_aov_hnb_iscv)
depth_aov_nb_iscv = upper_bound_pvalue(depth_aov_nb_iscv)
depth_aov_hp_iscv = upper_bound_pvalue(depth_aov_hp_iscv)
depth_aov_pois_iscv = upper_bound_pvalue(depth_aov_pois_iscv)

lp_aov_hnb_iscv = upper_bound_pvalue(lp_aov_hnb_iscv)

```

```

lp_aov_nb_iscv    = upper_bound_pvalue(lp_aov_nb_iscv)
lp_aov_hp_iscv    = upper_bound_pvalue(lp_aov_hp_iscv)
lp_aov_pois_iscv  = upper_bound_pvalue(lp_aov_pois_iscv)

chisq_hnb_iscv_cap = upper_bound_pvalue(chisq_iscv_rpp_hnb)
chisq_nb_iscv_cap  = upper_bound_pvalue(chisq_iscv_rpp_nb)
chisq_hp_iscv_cap  = upper_bound_pvalue(chisq_iscv_rpp_hp)
chisq_pois_iscv_cap = upper_bound_pvalue(chisq_iscv_rpp_pois)

```

4.5 Inspecting Trends of Z-Residuals for Checking Functional Forms

We next examine whether the linear covariate specification is adequate. For the linear HNB model, the posterior Z-residuals plotted against `depth` show a clear U-shaped pattern (?@fig-zresid-2a), and the grouped boxplots show systematic location changes across depth intervals (?@fig-zresid-2b). The grouped AOV diagnostic therefore targets the residual mean differences across depth groups, rather than only checking the marginal residual distribution. The replicated p -value histograms in ?@fig-zresid-2c-d show why this targeted diagnostic is useful: the SW test is mostly non-significant, while the depth-based AOV p -values are concentrated near zero. This indicates that the main deficiency is a nonlinear depth effect, not simply a global departure from normality.

```

par(mfrow = c(2,2))
xmax1=max(range(sw.test.zresid(Dolphins_zres_post_rpp_hnb),
                             aov.test.zresid(Dolphins_zres_post_rpp_hnb,X ="depth")))
xmin1=min(range(sw.test.zresid(Dolphins_zres_post_rpp_hnb),
                             aov.test.zresid(Dolphins_zres_post_rpp_hnb,X ="depth")))
plot (Dolphins_zres_post_rpp_hnb,irep = 7, x_axis_var="depth",
      normality.test=c("SW", "AOV", "BL"),
      main.title=expression("(a) HNB Linear, " * Z^"Post-RSP" * " Scatterplot"),
      cex.main=1.5,cex.lab=1.5, cex.axis=1.5,
      cex = 1.5)
lines(lowess(Dolphins$depth,Dolphins_zres_post_rpp_hnb[,2]),lwd = 2)
boxplot(Dolphins_zres_post_rpp_hnb,irep = 7,x_axis_var="depth",
        main.title=expression("(b) HNB Linear, " * Z^"Post-RSP" * " Boxplot"),
        cex.main=1.5,cex.lab=1.5, cex.axis=1.5)
hist(sw.test.zresid(Dolphins_zres_post_rpp_hnb),
     main=expression("(c) HNB Linear, " * Z^"Post-RSP"-SW * " P-values"),
     xlab=expression(Z^"Post-RSP"-SW * " Replicated P-values"),
     cex.main=1.4,cex.lab=1.5, cex.axis=1.5)

```

```

abline(v=upper_bound_pvalue(sw.test.zresid(Dolphins_zres_post_rpp_hnb)),col="red")

hist(aov.test.zresid(Dolphins_zres_post_rpp_hnb,X ="depth"),
     main=expression("(d) HNB Linear, " * Z^"Post-RSP"-AOV * " P-values"),
     breaks=20,
     cex.main=1.4,cex.lab=1.5, cex.axis=1.5,
     xlab=expression(Z^"Post-RSP"-AOV * " Replicated P-values"),xlim=c(xmin1,xmax1))
abline(v=upper_bound_pvalue(aov.test.zresid(Dolphins_zres_post_rpp_hnb,X ="depth")),col="red")

```

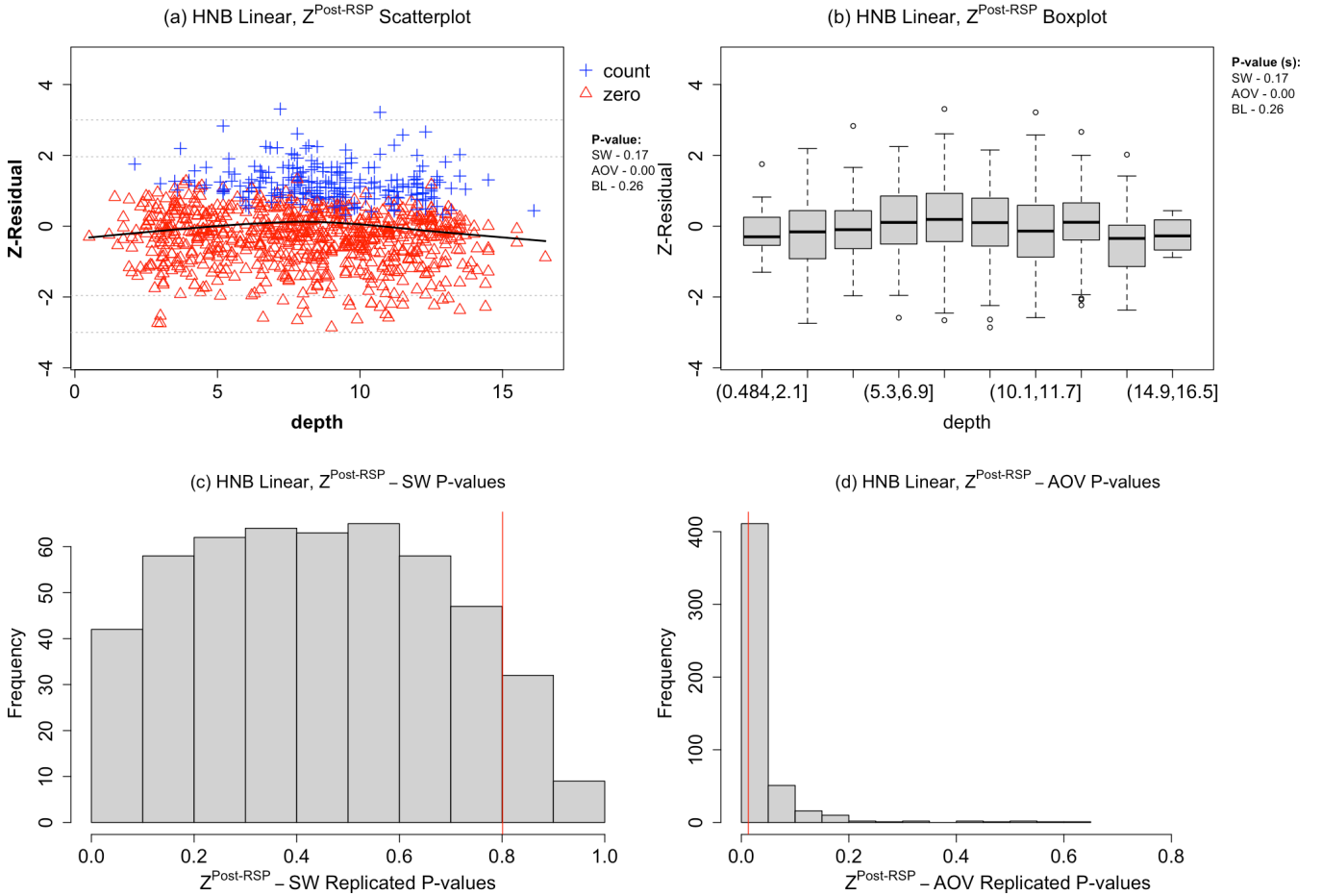


Figure 2: Posterior RSP Z-residual diagnostics for the linear HNB model. Panels (a) and (b) show Z-residuals against `depth` using a scatterplot and grouped boxplots; positive counts and zero observations are marked separately. Panels (c) and (d) show 500 replicated p -values for the $Z^{\text{Post-RSP}}$ -SW and depth-based $Z^{\text{Post-RSP}}$ -AOV tests; red lines mark the upper-bound summaries.

4.6 Z-residual Diagnostic Table for Linear Models

The table reports diagnostic p -values based on Z-residuals for four linear count models (HNB, NB, HP, Poisson), listing posterior Z-residual tests (Z-SW, Z-AOV with covariate `depth`, and $Z\text{-}\chi^2$ with effective d.f.).

```
library(dplyr)
library(gt)

df_linear <- data.frame(
  "Func.Form" = c("Linear", "Linear", "Linear", "Linear"),
  "Family"     = c("HNB", "NB", "HP", "Pois"),

  # Posterior Z-residual tests
  Z_SW_post      = c(pv_hnb, pv_nb, pv_hp, pv_pois),
  Z_AOV_lp_post  = c(lp_aov_hnb, lp_aov_nb, lp_aov_hp, lp_aov_pois),
  Z_AOV_depth_post = c(depth_aov_hnb, depth_aov_nb, depth_aov_hp, depth_aov_pois),
  Z_Chisq_post   = c(chisq_hnb_cap, chisq_nb_cap, chisq_hp_cap, chisq_pois_cap),

  # ISCV Z-residual tests
  Z_SW_iscv      = c(pv_hnb_iscv, pv_nb_iscv, pv_hp_iscv, pv_pois_iscv),
  Z_AOV_lp_iscv  = c(lp_aov_hnb_iscv, lp_aov_nb_iscv, lp_aov_hp_iscv, lp_aov_pois_iscv),
  Z_AOV_depth_iscv = c(depth_aov_hnb_iscv, depth_aov_nb_iscv, depth_aov_hp_iscv, depth_aov_pois_iscv),
  Z_Chisq_iscv   = c(chisq_hnb_iscv_cap, chisq_nb_iscv_cap, chisq_hp_iscv_cap, chisq_pois_iscv_cap),
)

df_linear %>%
  mutate(`Func.Form` = ifelse(dplyr::row_number() > 1, "", `Func.Form`)) %>%
  gt() %>%
  cols_label(
    Func.Form = "Func. Form",
    Family     = "Family",

    Z_SW_post  = "Z-SW",
    Z_AOV_lp_post = "Z-AOV-lp",
    Z_AOV_depth_post = "Z-AOV-depth",
    Z_Chisq_post = md("$Z\\text{-}\\chi^2$"),

    Z_SW_iscv  = "Z-SW",
    Z_AOV_lp_iscv = "Z-AOV-lp",
```

Table 1: Z-residual diagnostic p-values for the four linear count models. Posterior and ISCV entries are upper-bound summaries of 500 replicated p-values. The AOV diagnostics are conducted against both the fitted linear predictor (`lp`) and water depth (`depth`).

Func. Form	Family	Posterior Z-Residual Tests				ISCV Z-Residual Tests			
		Z-SW	Z-AOV-lp	Z-AOV-depth	$Z-\chi^2$	Z-SW	Z-AOV-lp	Z-AOV-depth	$Z-\chi^2$
Linear	HNB	0.801	0.914	0.013	0.812	0.449	0.985	0.015	0.675
	NB	0.326	0.029	0.012	0.923	0.895	0.023	0.009	0.742
	HP	0.000	0.766	0.022	0.000	0.000	0.759	0.020	0.000
	Pois	0.000	0.503	0.015	0.000	0.000	0.466	0.015	0.000

```

Z_AOV_depth_iscv = "Z-AOV-depth",
Z_Chisq_iscv      = md("$Z\\text{-}\\chi^2$")
) %>%
tab_spanner(
  label = "Posterior Z-Residual Tests",
  columns = c(Z_SW_post, Z_AOV_lp_post, Z_AOV_depth_post, Z_Chisq_post)
) %>%
tab_spanner(
  label = "ISCV Z-Residual Tests",
  columns = c(Z_SW_iscv, Z_AOV_lp_iscv, Z_AOV_depth_iscv, Z_Chisq_iscv)
) %>%
fmt_number(
  columns = c(
    Z_SW_post, Z_AOV_lp_post, Z_AOV_depth_post, Z_Chisq_post,
    Z_SW_iscv, Z_AOV_lp_iscv, Z_AOV_depth_iscv, Z_Chisq_iscv
  ),
  decimals = 3
) %>%
cols_width(`Func.Form` ~ px(95), Family ~ px(70)) %>%
cols_align(align = "left", columns = c(`Func.Form`, Family))

```

4.7 PPC and HPC Helper Functions

The next helper functions extract the main values returned by `ppc()` and `hpc()` into rectangular tables. These summaries include Pearson chi-square checks on the data scale and Z-residual-based SW and AOV checks.

```

.null_or <- function(x, y) {
  if (is.null(x) || length(x) == 0L) y else x
}

.get_named_aov_value <- function(obj, target) {
  vals <- obj$aov_rpp_all

  if (is.null(vals) || length(vals) == 0L) {
    vals <- obj$aov_rpp
  }

  if (is.null(vals) || length(vals) == 0L || all(is.na(vals))) {
    return(NA_real_)
  }

  nm <- names(vals)
  if (is.null(nm)) {
    if (length(vals) == 1L && identical(target, "lp")) {
      return(unname(as.numeric(vals[1L])))
    }
    return(NA_real_)
  }

  exact_names <- c(
    target,
    paste0("AOV[", target, "]"),
    paste0("Z-AOV[", target, "]")
  )

  idx <- which(nm %in% exact_names)
  if (length(idx) == 0L) {
    idx <- grep(paste0("\\[", target, "\\]"), nm)
  }
  if (length(idx) == 0L) {
    idx <- grep(paste0("(^[[:alnum:]])", target, "([[:alnum:]]|$)"), nm)
  }
  if (length(idx) == 0L) {
    return(NA_real_)
  }
}

```

```

  unname(as.numeric(vals[idx[1L]]))
}

.extract_predcheck_values <- function(obj) {
  c(
    Chisq = as.numeric(.null_or(obj$chisq, NA_real_)),
    Z_SW_randomized_mean = as.numeric(.null_or(obj$sw_rpp, .null_or(obj$mean_sw_pv, NA_real_))),
    Z_SW_midpoint = as.numeric(.null_or(obj$sw_mpp, NA_real_)),
    Z_AOV_lp = .get_named_aov_value(obj, "lp"),
    Z_AOV_depth = .get_named_aov_value(obj, "depth")
  )
}

.make_predcheck_table <- function(func_form, family, ppc_list, hpc_list) {
  ppc_vals <- t(vapply(ppc_list, .extract_predcheck_values, numeric(5L)))
  hpc_vals <- t(vapply(hpc_list, .extract_predcheck_values, numeric(5L)))

  colnames(ppc_vals) <- paste0("PPC_", colnames(ppc_vals))
  colnames(hpc_vals) <- paste0("HPC_", colnames(hpc_vals))

  data.frame(
    Func.Form = func_form,
    Family = family,
    ppc_vals,
    hpc_vals,
    check.names = FALSE
  )
}

.render_predcheck_gt <- function(tab) {
  tab %>%
  gt() %>%
  cols_label(
    Func.Form = "Func. Form",
    Family = "Family",
    PPC_Chisq = md("$\\chi^2$"),
    PPC_Z_SW_randomized_mean = "Z-SW RSP mean",
    PPC_Z_SW_midpoint = "Z-SW MPP",

```



```

PPC_Z_AOV_lp = "Z-AOV[lp]",
PPC_Z_AOV_depth = "Z-AOV[depth]",
HPC_Chisq = md("$\\chi^2$"),
HPC_Z_SW_randomized_mean = "Z-SW RSP mean",
HPC_Z_SW_midpoint = "Z-SW MPP",
HPC_Z_AOV_lp = "Z-AOV[lp]",
HPC_Z_AOV_depth = "Z-AOV[depth]"
) %>%
tab_spanner(
  label = "PPC",
  columns = c(PPC_Chisq, PPC_Z_SW_randomized_mean, PPC_Z_SW_midpoint,
    PPC_Z_AOV_lp, PPC_Z_AOV_depth)
) %>%
tab_spanner(
  label = "HPC",
  columns = c(HPC_Chisq, HPC_Z_SW_randomized_mean, HPC_Z_SW_midpoint,
    HPC_Z_AOV_lp, HPC_Z_AOV_depth)
) %>%
fmt_number(
  columns = c(PPC_Chisq, PPC_Z_SW_randomized_mean, PPC_Z_SW_midpoint,
    PPC_Z_AOV_lp, PPC_Z_AOV_depth,
    HPC_Chisq, HPC_Z_SW_randomized_mean, HPC_Z_SW_midpoint,
    HPC_Z_AOV_lp, HPC_Z_AOV_depth),
  decimals = 3
) %>%
cols_width(Func.Form ~ px(95), Family ~ px(70)) %>%
cols_align(align = "left", columns = c(Func.Form, Family)) %>%
tab_options(table.align = "center")
}

```

4.8 PPC

```

ndraws_use <- 4000
seed_use <- 123
x_check <- c("lp", "depth")

ppc_linear_file <- file.path(resources_path, "ppc_linear_list_all.rds")

```

```

if (!force_recompute_ppc_hpc && file.exists(ppc_linear_file)) {
  ppc_linear_list <- readRDS(ppc_linear_file)
} else {
  ppc_linear_list <- list(
    HNB = ppc(fit = Dolphins_hnb, data = Dolphins, x = x_check,
              ndraws = ndraws_use, seed = seed_use, randomized = TRUE),
    NB = ppc(fit = Dolphins_nb, data = Dolphins, x = x_check,
              ndraws = ndraws_use, seed = seed_use, randomized = TRUE),
    HP = ppc(fit = Dolphins_hp, data = Dolphins, x = x_check,
              ndraws = ndraws_use, seed = seed_use, randomized = TRUE),
    Poisson = ppc(fit = Dolphins_pois, data = Dolphins, x = x_check,
                   ndraws = ndraws_use, seed = seed_use, randomized = TRUE)
  )

  saveRDS(ppc_linear_list, ppc_linear_file)
}

ppc_hnb <- ppc_linear_list$HNB
ppc_nb <- ppc_linear_list$NB
ppc_hp <- ppc_linear_list$HP
ppc_pois <- ppc_linear_list$Poisson

```

4.9 HPC

```

filename_hpc_linear <- file.path(resources_path, "fit_Dolphins_linear_train_for_hpc.Rdata")
hpc_linear_file <- file.path(resources_path, "hpc_linear_list_all.rds")

if (!forcerefit && file.exists(filename_hpc_linear)) {
  message("Loading linear training-data fitted models from: ", filename_hpc_linear)
  load(filename_hpc_linear)
} else {
  message("Creating holdout split, fitting linear training-data models, and saving to: ", filename_hpc_linear)

  set.seed(20260514)

  # Stratified holdout split so that the zero proportion is roughly preserved.

```

```

zero_id <- which(Dolphins$sightings == 0)
pos_id  <- which(Dolphins$sightings > 0)

hpc_id <- c(
  sample(zero_id, size = floor(0.25 * length(zero_id))),
  sample(pos_id,  size = floor(0.25 * length(pos_id)))
)
hpc_id <- sort(hpc_id)

Dolphins_train <- Dolphins[-hpc_id, , drop = FALSE]
Dolphins_hpc   <- Dolphins[ hpc_id, , drop = FALSE]

Dolphins_count_formula_linear <- sightings ~ depth + temp + dr + salinity + turbidity + year
Dolphins_zero_formula_linear  <- hu ~ depth + temp + dr + salinity + turbidity + year

Dolphins_hnb_train <- brm(
  bf(Dolphins_count_formula_linear, Dolphins_zero_formula_linear),
  data      = Dolphins_train,
  family    = hurdle_negbinomial(link = "log", link_shape = "identity", link_hu = "logit"),
  save_pars = save_pars(all = TRUE),
  refresh   = 0
)

Dolphins_hp_train <- brm(
  bf(Dolphins_count_formula_linear, Dolphins_zero_formula_linear),
  data      = Dolphins_train,
  family    = hurdle_poisson(link = "log", link_hu = "logit"),
  save_pars = save_pars(all = TRUE),
  refresh   = 0
)

Dolphins_nb_train <- brm(
  Dolphins_count_formula_linear,
  data      = Dolphins_train,
  family    = brms::negbinomial(link = "log", link_shape = "identity"),
  save_pars = save_pars(all = TRUE),
  refresh   = 0
)

```

```

Dolphins_pois_train <- brm(
  Dolphins_count_formula_linear,
  data = Dolphins_train,
  family = poisson(link = "log"),
  save_pars = save_pars(all = TRUE),
  refresh = 0
)

save(
  Dolphins_train, Dolphins_hpc,
  Dolphins_hnb_train, Dolphins_hp_train, Dolphins_nb_train, Dolphins_pois_train,
  file = filename_hpc_linear
)
}

if (!force_recompute_ppc_hpc && file.exists(hpc_linear_file)) {
  hpc_linear_list <- readRDS(hpc_linear_file)
} else {
  hpc_linear_list <- list(
    HNB = hpc(fit = Dolphins_hnb_train, newdata = Dolphins_hpc, x = x_check,
              ndraws = ndraws_use, seed = seed_use, randomized = TRUE),
    NB = hpc(fit = Dolphins_nb_train, newdata = Dolphins_hpc, x = x_check,
             ndraws = ndraws_use, seed = seed_use, randomized = TRUE),
    HP = hpc(fit = Dolphins_hp_train, newdata = Dolphins_hpc, x = x_check,
             ndraws = ndraws_use, seed = seed_use, randomized = TRUE),
    Poisson = hpc(fit = Dolphins_pois_train, newdata = Dolphins_hpc, x = x_check,
                  ndraws = ndraws_use, seed = seed_use, randomized = TRUE)
  )

  saveRDS(hpc_linear_list, hpc_linear_file)
}

hpc_hnb <- hpc_linear_list$HNB
hpc_nb <- hpc_linear_list$NB
hpc_hp <- hpc_linear_list$HP
hpc_pois <- hpc_linear_list$Poisson

```

4.10 PPC and HPC Diagnostic Table

```
predcheck_linear_table <- data.frame(
  `Func.Form` = c("Linear", "Linear", "Linear", "Linear"),
  Family = c("HNB", "NB", "HP", "Pois"),

  PPC_Chisq = c(ppc_hnb$chisq, ppc_nb$chisq, ppc_hp$chisq, ppc_pois$chisq),
  PPC_Z_SW_randomized_mean = c(ppc_hnb$sw_rpp, ppc_nb$sw_rpp, ppc_hp$sw_rpp, ppc_pois$sw_rpp),
  PPC_Z_SW_midpoint = c(ppc_hnb$sw_mpp, ppc_nb$sw_mpp, ppc_hp$sw_mpp, ppc_pois$sw_mpp),
  PPC_Z_AOV_lp = c(ppc_hnb$aov_rpp_all[["AOV[lp]"]], ppc_nb$aov_rpp_all[["AOV[lp]"]], ppc_hp$aov_rpp_all[["AOV[lp]"]], ppc_pois$aov_rpp_all[["AOV[lp]"]]),
  PPC_Z_AOV_depth = c(ppc_hnb$aov_rpp_all[["AOV[depth]"]], ppc_nb$aov_rpp_all[["AOV[depth]"]], ppc_hp$aov_rpp_all[["AOV[depth]"]], ppc_pois$aov_rpp_all[["AOV[depth]"]]),

  HPC_Chisq = c(hpc_hnb$chisq, hpc_nb$chisq, hpc_hp$chisq, hpc_pois$chisq),
  HPC_Z_SW_randomized_mean = c(hpc_hnb$sw_rpp, hpc_nb$sw_rpp, hpc_hp$sw_rpp, hpc_pois$sw_rpp),
  HPC_Z_SW_midpoint = c(hpc_hnb$sw_mpp, hpc_nb$sw_mpp, hpc_hp$sw_mpp, hpc_pois$sw_mpp),
  HPC_Z_AOV_lp = c(hpc_hnb$aov_rpp_all[["AOV[lp]"]], hpc_nb$aov_rpp_all[["AOV[lp]"]], hpc_hp$aov_rpp_all[["AOV[lp]"]], hpc_pois$aov_rpp_all[["AOV[lp]"]]),
  HPC_Z_AOV_depth = c(hpc_hnb$aov_rpp_all[["AOV[depth]"]], hpc_nb$aov_rpp_all[["AOV[depth]"]], hpc_hp$aov_rpp_all[["AOV[depth]"]], hpc_pois$aov_rpp_all[["AOV[depth]"]]),
  check.names = FALSE
)

predcheck_linear_table %>%
  mutate(`Func.Form` = ifelse(dplyr::row_number() > 1, "", `Func.Form`)) %>%
  gt() %>%
  cols_label(
    `Func.Form` = "Func. Form",
    Family = "Family",
    PPC_Chisq = md("$\\chi^2$"),
    PPC_Z_SW_randomized_mean = "Z-SW RSP mean",
    PPC_Z_SW_midpoint = "Z-SW MPP",
    PPC_Z_AOV_lp = "Z-AOV-lp",
    PPC_Z_AOV_depth = "Z-AOV-depth",
    HPC_Chisq = md("$\\chi^2$"),
    HPC_Z_SW_randomized_mean = "Z-SW RSP mean",
    HPC_Z_SW_midpoint = "Z-SW MPP",
    HPC_Z_AOV_lp = "Z-AOV-lp",
    HPC_Z_AOV_depth = "Z-AOV-depth"
  ) %>%
  tab_spanner(
    label = "PPC",
```

Table 2: PPC diagnostic p-values from full-data fits and HPC diagnostic p-values from training-data fits evaluated on the held-out data for the four linear models.

Func. Form	Family	PPC							
		χ^2	Z-SW	RSP mean	Z-SW MPP	Z-AOV-lp	Z-AOV-depth	χ^2	Z-SW R
Linear	HNB	0.016		0.301	0.285	0.419	0.033	0.333	
	NB	0.531		0.387	0.475	0.152	0.025	0.884	
	HP	0.001		0.000	0.000	0.455	0.041	0.419	
	Pois	0.000		0.000	0.000	0.300	0.014	0.000	

```

columns = c(PPC_Chisq, PPC_Z_SW_randomized_mean, PPC_Z_SW_midpoint,
             PPC_Z_AOV_lp, PPC_Z_AOV_depth)
) %>%
tab_spanner(
  label = "HPC",
  columns = c(HPC_Chisq, HPC_Z_SW_randomized_mean, HPC_Z_SW_midpoint,
              HPC_Z_AOV_lp, HPC_Z_AOV_depth)
) %>%
fmt_number(
  columns = c(PPC_Chisq, PPC_Z_SW_randomized_mean, PPC_Z_SW_midpoint,
              PPC_Z_AOV_lp, PPC_Z_AOV_depth,
              HPC_Chisq, HPC_Z_SW_randomized_mean, HPC_Z_SW_midpoint,
              HPC_Z_AOV_lp, HPC_Z_AOV_depth),
  decimals = 3
) %>%
cols_width(`Func.Form` ~ px(95), Family ~ px(70)) %>%
cols_align(align = "left", columns = c(`Func.Form`, Family)) %>%
tab_options(table.align = "center")

```

5 Diagnosing HNB and NB Nonlinear Models

5.1 Fitting Two Nonlinear Models with a Piecewise Quadratic Function for Depth

The linear HNB diagnostics suggest that a simple linear effect of `depth` is inadequate. To demonstrate one possible refinement, we use a piecewise quadratic effect with a turning point at `depth = 7.95`, motivated

by the residual pattern in [?@fig-zresid-2a](#). We create two variables:

$$\text{below7.95} = \{(\text{depth} - 7.95)^2\} 1(\text{depth} < 7.95)$$

$$\text{above7.95} = \{(\text{depth} - 7.95)^2\} 1(\text{depth} \geq 7.95).$$

```
filename <- file.path(resources_path, "fit_Dolphins_n12.Rdata")
# Set forcerefit to TRUE if you want to override the saved file
if (!exists("forcerefit")) forcerefit <- FALSE
if (forcerefit || !file.exists(filename)) {
  Dolphins$below8 <- ifelse(Dolphins$depth < 7.95, (Dolphins$depth - 7.95)^2, 0)
  Dolphins$above8 <- ifelse(Dolphins$depth >= 7.95, (Dolphins$depth - 7.95)^2, 0)

  Dolphins_count_formula_n12 <- formula(paste("sightings ~ below8 + above8 + temp + dr + salinity + turbidity"))
  Dolphins_zero_formula_n12 <- formula(paste("hu ~ below8 + above8 + temp + dr + salinity + turbidity"))

  Dolphins_hnb_n12 <- brm(
    bf(Dolphins_count_formula_n12, Dolphins_zero_formula_n12),
    data = Dolphins,
    family = hurdle_negbinomial(link = "log", link_shape = "identity", link_hu = "logit"),
    save_pars = save_pars(all = TRUE),
    refresh = 0
  )

  Dolphins_nb_n12 <- brm(
    Dolphins_count_formula_n12,
    data = Dolphins,
    family = brms::negbinomial(link = "log", link_shape = "identity"),
    save_pars = save_pars(all = TRUE),
    refresh = 0
  )
  save(Dolphins_hnb_n12, Dolphins_nb_n12, file = filename)
} else{
  base::load(filename)
}
Dolphins$below8 <- ifelse(Dolphins$depth < 7.95, (Dolphins$depth - 7.95)^2, 0)
Dolphins$above8 <- ifelse(Dolphins$depth >= 7.95, (Dolphins$depth - 7.95)^2, 0)
```

5.2 Computing Z-Residuals

After refitting the HNB and NB models with the piecewise quadratic depth terms, we recompute posterior and ISCV RSP Z-residuals.

```
zres_nonlinear_post_file <- file.path(resources_path, "zres_nonlinear_post_list_all.rds")
zres_nonlinear_iscv_file <- file.path(resources_path, "zres_nonlinear_iscv_list_all.rds")

required_nonlinear_zres_names <- c("HNB", "NB")

if (
  !force_recompute_zres &&
  file.exists(zres_nonlinear_post_file) &&
  file.exists(zres_nonlinear_iscv_file)
) {
  zres_nonlinear_post_list <- readRDS(zres_nonlinear_post_file)
  zres_nonlinear_iscv_list <- readRDS(zres_nonlinear_iscv_file)

  missing_post <- setdiff(required_nonlinear_zres_names, names(zres_nonlinear_post_list))
  missing_iscv <- setdiff(required_nonlinear_zres_names, names(zres_nonlinear_iscv_list))

  if (length(missing_post) || length(missing_iscv)) {
    stop(
      "Saved nonlinear Z-residual files do not contain the expected model names.\n",
      "Missing from posterior list: ", paste(missing_post, collapse = ", "), "\n",
      "Missing from ISCV list: ", paste(missing_iscv, collapse = ", "),
      call. = FALSE
    )
  }
} else {
  zres_nonlinear_post_list <- list(
    HNB = Zresidual(Dolphins_hnb_nl2, data = Dolphins, mcmc_summarize = "post", nrep = nrep_z),
    NB = Zresidual(Dolphins_nb_nl2, data = Dolphins, mcmc_summarize = "post", nrep = nrep_z)
  )

  zres_nonlinear_iscv_list <- list(
    HNB = Zresidual(Dolphins_hnb_nl2, data = Dolphins, mcmc_summarize = "iscv", nrep = nrep_z),
    NB = Zresidual(Dolphins_nb_nl2, data = Dolphins, mcmc_summarize = "iscv", nrep = nrep_z)
  )
}
```



```

saveRDS(zres_nonlinear_post_list, zres_nonlinear_post_file)
saveRDS(zres_nonlinear_iscv_list, zres_nonlinear_iscv_file)
}

Dolphins_zres_post_rpp_hnb_nl2 <- zres_nonlinear_post_list$HNB
Dolphins_zres_post_rpp_nb_nl2  <- zres_nonlinear_post_list$NB
Dolphins_zres_iscv_rpp_hnb_nl2 <- zres_nonlinear_iscv_list$HNB
Dolphins_zres_iscv_rpp_nb_nl2  <- zres_nonlinear_iscv_list$NB

```

5.3 Diagnostic Tests

We repeat the linear-model diagnostic pipeline for the nonlinear specifications (nonlinear HNB and NB models). For each posterior draw we compute Z-residuals and run SW normality tests, ANOVA of Z-residuals against the covariate **depth** using binned fitted values with safeguards for sparse bins, and an omnibus Z-chi-squared test with effective degrees of freedom obtained from PSIS-LOO.

```

pv_hnb_nl2 <- sw.test.zresid(Dolphins_zres_post_rpp_hnb_nl2)
pv_nb_nl2  <- sw.test.zresid(Dolphins_zres_post_rpp_nb_nl2)

pv_hnb_iscv_nl2 <- sw.test.zresid(Dolphins_zres_iscv_rpp_hnb_nl2)
pv_nb_iscv_nl2  <- sw.test.zresid(Dolphins_zres_iscv_rpp_nb_nl2)

aov_depth_hnb_nl2 <- aov.test.zresid(Dolphins_zres_post_rpp_hnb_nl2, X = "depth")
aov_depth_nb_nl2  <- aov.test.zresid(Dolphins_zres_post_rpp_nb_nl2, X = "depth")

aov_depth_hnb_iscv_nl2 <- aov.test.zresid(Dolphins_zres_iscv_rpp_hnb_nl2, X = "depth")
aov_depth_nb_iscv_nl2  <- aov.test.zresid(Dolphins_zres_iscv_rpp_nb_nl2, X = "depth")

aov_lp_hnb_nl2 <- aov.test.zresid(Dolphins_zres_post_rpp_hnb_nl2, X = "lp")
aov_lp_nb_nl2  <- aov.test.zresid(Dolphins_zres_post_rpp_nb_nl2, X = "lp")

aov_lp_hnb_iscv_nl2 <- aov.test.zresid(Dolphins_zres_iscv_rpp_hnb_nl2, X = "lp")
aov_lp_nb_iscv_nl2  <- aov.test.zresid(Dolphins_zres_iscv_rpp_nb_nl2, X = "lp")

p_effective_hnb_nl2 <- loo(Dolphins_hnb_nl2, moment_match = TRUE)$estimates["p_loo", "Estimate"]
p_effective_nb_nl2  <- loo(Dolphins_nb_nl2,  moment_match = TRUE)$estimates["p_loo", "Estimate"]

n_obs <- nrow(Dolphins_zres_post_rpp_hnb_nl2)

```

```

ncols <- min(500L, ncol(Dolphins_zres_post_rpp_hnb_nl2), ncol(Dolphins_zres_post_rpp_nb_nl2),
            ncol(Dolphins_zres_iscv_rpp_hnb_nl2), ncol(Dolphins_zres_iscv_rpp_nb_nl2))

chisq_zres_post_rpp_hnb_nl2 <- numeric(ncols)
chisq_zres_post_rpp_nb_nl2  <- numeric(ncols)
chisq_post_rpp_hnb_nl2     <- numeric(ncols)
chisq_post_rpp_nb_nl2      <- numeric(ncols)

chisq_zres_iscv_rpp_hnb_nl2 <- numeric(ncols)
chisq_zres_iscv_rpp_nb_nl2  <- numeric(ncols)
chisq_iscv_rpp_hnb_nl2     <- numeric(ncols)
chisq_iscv_rpp_nb_nl2      <- numeric(ncols)

for (i in seq_len(ncols)) {
  chisq_zres_post_rpp_hnb_nl2[i] <- sum(Dolphins_zres_post_rpp_hnb_nl2[, i]^2, na.rm = TRUE)
  chisq_zres_post_rpp_nb_nl2[i]  <- sum(Dolphins_zres_post_rpp_nb_nl2[, i]^2, na.rm = TRUE)

  chisq_zres_iscv_rpp_hnb_nl2[i] <- sum(Dolphins_zres_iscv_rpp_hnb_nl2[, i]^2, na.rm = TRUE)
  chisq_zres_iscv_rpp_nb_nl2[i]  <- sum(Dolphins_zres_iscv_rpp_nb_nl2[, i]^2, na.rm = TRUE)

  chisq_post_rpp_hnb_nl2[i] <- pchisq(q = chisq_zres_post_rpp_hnb_nl2[i], df = n_obs - p_effective)
  chisq_post_rpp_nb_nl2[i]  <- pchisq(q = chisq_zres_post_rpp_nb_nl2[i], df = n_obs - p_effective)

  chisq_iscv_rpp_hnb_nl2[i] <- pchisq(q = chisq_zres_iscv_rpp_hnb_nl2[i], df = n_obs, lower.tail = FALSE)
  chisq_iscv_rpp_nb_nl2[i]  <- pchisq(q = chisq_zres_iscv_rpp_nb_nl2[i], df = n_obs, lower.tail = FALSE)
}

```

5.4 Inspecting Trends of Z-Residuals for Checking Functional Forms

In Figure 3, the residual scatterplot and grouped boxplots for the nonlinear HNB model no longer show the strong U-shaped depth pattern seen in the linear fit. The replicated p -value histograms for both $Z^{\text{Post-RSP-SW}}$ and the depth-based $Z^{\text{Post-RSP-AOV}}$ test are mostly away from zero, supporting the adequacy of the nonlinear depth specification for this candidate model.

```

par(mfrow = c(2,2))
xmax1=max(range(sw.test.zresid(Dolphins_zres_post_rpp_hnb_nl2),aov_depth_hnb_nl2))
xmin1=min(range(sw.test.zresid(Dolphins_zres_post_rpp_hnb_nl2),aov_depth_hnb_nl2))
plot (Dolphins_zres_post_rpp_hnb_nl2,irep = 8,

```

```

    x_axis_var=Dolphins$depth,normality.test=c("SW", "AOV", "BL"),
    main.title=expression("(a) HNB Nonlinear, " * Z^"Post-RSP" * " Scatterplot"),
    xlab="depth",
    cex.main=1.5,cex.lab=1.5, cex.axis=1.5
    #,cex = 1.5)
)
lines(lowess(Dolphins$depth,Dolphins_zres_post_rpp_hnb_nl2[,8]),lwd = 2)
boxplot(Dolphins_zres_post_rpp_hnb_nl2,irep = 2,x_axis_var=Dolphins$depth,
    main.title=expression("(b) HNB Nonlinear, " * Z^"Post-RSP" * " Boxplot"),
    xlab="depth",
    cex.main=1.5,cex.lab=1.5, cex.axis=1.5
    # ,cex = 1.5)
)
hist(sw.test.zresid(Dolphins_zres_post_rpp_hnb_nl2),
    main=expression("(c) HNB Nonlinear, " * Z^"Post-RSP"-SW * " P-values"),
    xlab=expression(Z^"Post-RSP"-SW * " Replicated P-values"),xlim=c(xmin1,xmax1),
    cex.main=1.4,cex.lab=1.5, cex.axis=1.5)
abline(v=upper_bound_pvalue(sw.test.zresid(Dolphins_zres_post_rpp_hnb_nl2)),col="red")

hist(aov_depth_hnb_nl2,
    main=expression("(d) HNB Nonlinear, " * Z^"Post-RSP"-AOV * " P-values"),
    breaks=20,
    xlab=expression(Z^"Post-RSP"-AOV * " Replicated P-values"),xlim=c(xmin1,xmax1),
    cex.main=1.4,cex.lab=1.5, cex.axis=1.5)
abline(v=upper_bound_pvalue(aov_depth_hnb_nl2),col="red")

```

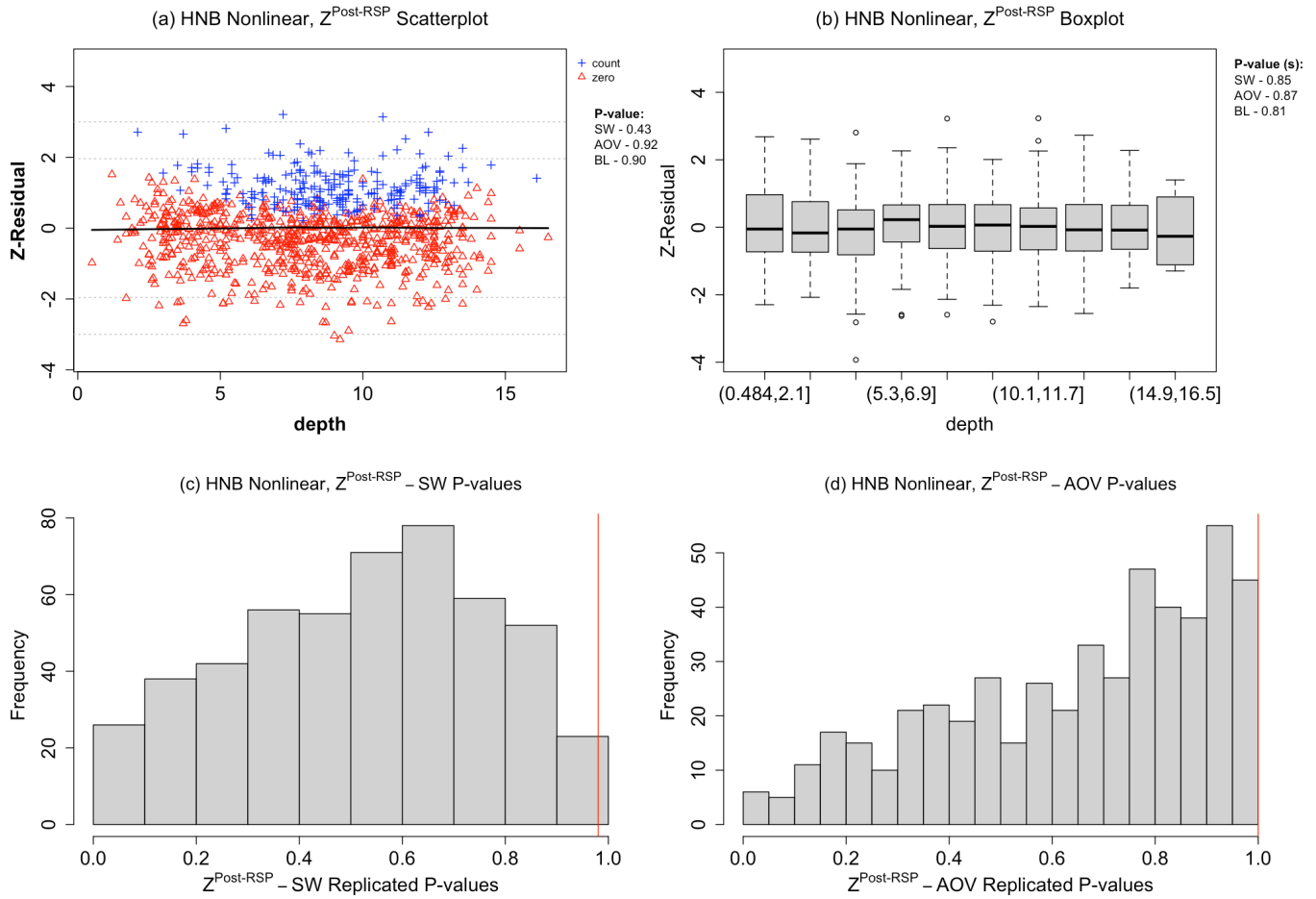


Figure 3: Posterior RSP Z-residual diagnostics for the nonlinear HNB model. Panels (a) and (b) show a scatterplot and grouped boxplots of Z-residuals against `depth`, distinguishing positive count observations from zero observations. Panels (c) and (d) display histograms of 500 replicated p -values for $Z^{\text{Post-RSP}}$ -SW and the depth-based $Z^{\text{Post-RSP}}$ -AOV test.

5.5 Z-residual Diagnostic Table for Nonlinear Models

Table 3 reports the same Z-residual diagnostics for the two nonlinear models. These values can be compared with the linear-model diagnostics to assess whether the nonlinear depth specification removes the residual pattern.

```
library(dplyr)
library(gt)

pv_hnb_n12_cap <- upper_bound_pvalue(pv_hnb_n12)
pv_nb_n12_cap <- upper_bound_pvalue(pv_nb_n12)

aov_depth_hnb_n1 <- upper_bound_pvalue(aov_depth_hnb_n12)
```

```

aov_depth_nb_nl <- upper_bound_pvalue(aov_depth_nb_nl2)

aov_lp_hnb_nl <- upper_bound_pvalue(aov_lp_hnb_nl2)
aov_lp_nb_nl <- upper_bound_pvalue(aov_lp_nb_nl2)

chisq_hnb_nl <- upper_bound_pvalue(chisq_post_rpp_hnb_nl2)
chisq_nb_nl <- upper_bound_pvalue(chisq_post_rpp_nb_nl2)

# ISCV
pv_hnb_nl2_iscv_cap <- upper_bound_pvalue(pv_hnb_iscv_nl2)
pv_nb_nl2_iscv_cap <- upper_bound_pvalue(pv_nb_iscv_nl2)

aov_depth_hnb_nl_iscv <- upper_bound_pvalue(aov_depth_hnb_iscv_nl2)
aov_depth_nb_nl_iscv <- upper_bound_pvalue(aov_depth_nb_iscv_nl2)

aov_lp_hnb_nl_iscv <- upper_bound_pvalue(aov_lp_hnb_iscv_nl2)
aov_lp_nb_nl_iscv <- upper_bound_pvalue(aov_lp_nb_iscv_nl2)

chisq_hnb_nl_iscv <- upper_bound_pvalue(chisq_iscv_rpp_hnb_nl2)
chisq_nb_nl_iscv <- upper_bound_pvalue(chisq_iscv_rpp_nb_nl2)

results_table_nl <- data.frame(
  "Func.Form"      = c("Nonlinear", "Nonlinear"),
  "Family"         = c("HNB", "NB"),

  # Posterior
  Z_SW_post        = c(pv_hnb_nl2_cap, pv_nb_nl2_cap),
  Z_AOV_lp_post    = c(aov_lp_hnb_nl, aov_lp_nb_nl),
  Z_AOV_post       = c(aov_depth_hnb_nl, aov_depth_nb_nl),
  Z_Chisq_post     = c(chisq_hnb_nl, chisq_nb_nl),

  # ISCV
  Z_SW_iscv        = c(pv_hnb_nl2_iscv_cap, pv_nb_nl2_iscv_cap),
  Z_AOV_lp_iscv    = c(aov_lp_hnb_nl_iscv, aov_lp_nb_nl_iscv),
  Z_AOV_iscv       = c(aov_depth_hnb_nl_iscv, aov_depth_nb_nl_iscv),
  Z_Chisq_iscv     = c(chisq_hnb_nl_iscv, chisq_nb_nl_iscv)
)

results_table_nl %>%

```

```

mutate(`Func.Form` = ifelse(dplyr::row_number() > 1, "", `Func.Form`)) %>%
gt() %>%
cols_label(
  Func.Form      = "Func. Form",
  Family         = "Family",
  Z_SW_post      = "Z-SW",
  Z_AOV_lp_post  = "Z-AOV-lp",
  Z_AOV_post     = "Z-AOV-depth",
  Z_Chisq_post   = md("$Z\\text{-}\\chi^2$"),
  Z_SW_iscv      = "Z-SW",
  Z_AOV_lp_iscv  = "Z-AOV-lp",
  Z_AOV_iscv     = "Z-AOV-depth",
  Z_Chisq_iscv   = md("$Z\\text{-}\\chi^2$")
) %>%
tab_spanner(
  label = "Posterior Z-Residual Tests",
  columns = c(Z_SW_post, Z_AOV_lp_post, Z_AOV_post, Z_Chisq_post)
) %>%
tab_spanner(
  label = "ISCV Z-Residual Tests",
  columns = c(Z_SW_iscv, Z_AOV_lp_iscv, Z_AOV_iscv, Z_Chisq_iscv)
) %>%
fmt_number(
  columns = c(
    Z_SW_post, Z_AOV_lp_post, Z_AOV_post, Z_Chisq_post,
    Z_SW_iscv, Z_AOV_lp_iscv, Z_AOV_iscv, Z_Chisq_iscv
  ),
  decimals = 3
) %>%
cols_width(`Func.Form` ~ px(95), Family ~ px(70)) %>%
cols_align(align = "left", columns = c(`Func.Form`, Family))

```

5.6 PPC and HPC for Nonlinear Models

```

ndraws_use_nl <- ndraws_use
seed_use_nl <- seed_use
x_check_nl <- c("lp", "depth")

```

Table 3: Z-residual diagnostic p-values for the two nonlinear count models. Posterior and ISCV entries are upper-bound summaries of 500 replicated p-values. The AOV diagnostics are conducted against both the fitted linear predictor (`lp`) and water depth (`depth`).

Func. Form	Family	Posterior Z-Residual Tests				ISCV Z-Residual Tests			
		Z-SW	Z-AOV-lp	Z-AOV-depth	$Z-\chi^2$	Z-SW	Z-AOV-lp	Z-AOV-depth	$Z-\chi^2$
Nonlinear	HNB	0.916	0.294	1.000	0.730	0.554	0.293	1.000	0.563
	NB	0.311	0.709	0.633	0.876	0.027	0.788	0.635	0.571

```

ppc_nonlinear_file <- file.path(resources_path, "ppc_nonlinear_list_all.rds")
hpc_nonlinear_file <- file.path(resources_path, "hpc_nonlinear_list_all.rds")
filename_hpc_nl2 <- file.path(resources_path, "fit_Dolphins_nl2_train_for_hpc.Rdata")

# PPC uses the models fitted on the full dataset.
if (!force_recompute_ppc_hpc && file.exists(ppc_nonlinear_file)) {
  ppc_nonlinear_list <- readRDS(ppc_nonlinear_file)
} else {
  ppc_nonlinear_list <- list(
    HNB = ppc(fit = Dolphins_hnb_nl2, data = Dolphins, x = x_check_nl,
              ndraws = ndraws_use_nl, seed = seed_use_nl, randomized = TRUE),
    NB = ppc(fit = Dolphins_nb_nl2, data = Dolphins, x = x_check_nl,
             ndraws = ndraws_use_nl, seed = seed_use_nl, randomized = TRUE)
  )

  saveRDS(ppc_nonlinear_list, ppc_nonlinear_file)
}

ppc_hnb_nl2 <- ppc_nonlinear_list$HNB
ppc_nb_nl2 <- ppc_nonlinear_list$NB

# HPC uses nonlinear models refitted on the training data only.
if (!forcerefit && file.exists(filename_hpc_nl2)) {
  message("Loading nonlinear training-data fitted models from: ", filename_hpc_nl2)
  load(filename_hpc_nl2)
} else {
  message("Fitting nonlinear training-data models and saving to: ", filename_hpc_nl2)

```

```

Dolphins_train_nl <- Dolphins_train
Dolphins_hpc_nl   <- Dolphins_hpc

Dolphins_train_nl$below8 <- ifelse(
  Dolphins_train_nl$depth < 7.95,
  (Dolphins_train_nl$depth - 7.95)^2,
  0
)
Dolphins_train_nl$above8 <- ifelse(
  Dolphins_train_nl$depth >= 7.95,
  (Dolphins_train_nl$depth - 7.95)^2,
  0
)
Dolphins_hpc_nl$below8 <- ifelse(
  Dolphins_hpc_nl$depth < 7.95,
  (Dolphins_hpc_nl$depth - 7.95)^2,
  0
)
Dolphins_hpc_nl$above8 <- ifelse(
  Dolphins_hpc_nl$depth >= 7.95,
  (Dolphins_hpc_nl$depth - 7.95)^2,
  0
)

Dolphins_count_formula_nl2_hpc <- sightings ~ below8 + above8 + temp + dr + salinity + turbidity
Dolphins_zero_formula_nl2_hpc  <- hu ~ below8 + above8 + temp + dr + salinity + turbidity + year

Dolphins_hnb_nl2_train <- brm(
  bf(Dolphins_count_formula_nl2_hpc, Dolphins_zero_formula_nl2_hpc),
  data      = Dolphins_train_nl,
  family    = hurdle_negbinomial(link = "log", link_shape = "identity", link_hu = "logit"),
  save_pars = save_pars(all = TRUE),
  refresh   = 0
)

Dolphins_nb_nl2_train <- brm(
  Dolphins_count_formula_nl2_hpc,
  data      = Dolphins_train_nl,
  family    = brms::negbinomial(link = "log", link_shape = "identity"),

```



```

    save_pars = save_pars(all = TRUE),
    refresh   = 0
  )

  save(
    Dolphins_train_nl, Dolphins_hpc_nl,
    Dolphins_hnb_nl2_train, Dolphins_nb_nl2_train,
    file = filename_hpc_nl2
  )
}

if (!force_recompute_ppc_hpc && file.exists(hpc_nonlinear_file)) {
  hpc_nonlinear_list <- readRDS(hpc_nonlinear_file)
} else {
  hpc_nonlinear_list <- list(
    HNB = hpc(fit = Dolphins_hnb_nl2_train, newdata = Dolphins_hpc_nl,
              x = x_check_nl, ndraws = ndraws_use_nl,
              seed = seed_use_nl, randomized = TRUE),
    NB = hpc(fit = Dolphins_nb_nl2_train, newdata = Dolphins_hpc_nl,
             x = x_check_nl, ndraws = ndraws_use_nl,
             seed = seed_use_nl, randomized = TRUE)
  )

  saveRDS(hpc_nonlinear_list, hpc_nonlinear_file)
}

hpc_hnb_nl2 <- hpc_nonlinear_list$HNB
hpc_nb_nl2  <- hpc_nonlinear_list$NB

```

```

predcheck_nonlinear_table <- data.frame(
  `Func.Form` = c("Nonlinear", "Nonlinear"),
  Family = c("HNB", "NB"),

  PPC_Chisq = c(ppc_hnb_nl2$chisq, ppc_nb_nl2$chisq),
  PPC_Z_SW_randomized_mean = c(ppc_hnb_nl2$sw_rpp, ppc_nb_nl2$sw_rpp),
  PPC_Z_SW_midpoint = c(ppc_hnb_nl2$sw_mpp, ppc_nb_nl2$sw_mpp),
  PPC_Z_AOV_lp = c(ppc_hnb_nl2$aov_rpp_all[["AOV[lp]"]], ppc_nb_nl2$aov_rpp_all[["AOV[lp]"]]),
  PPC_Z_AOV_depth = c(ppc_hnb_nl2$aov_rpp_all[["AOV[depth]"]], ppc_nb_nl2$aov_rpp_all[["AOV[depth]"]])
)

```

```

HPC_Chisq = c(hpc_hnb_n12$chisq, hpc_nb_n12$chisq),
HPC_Z_SW_randomized_mean = c(hpc_hnb_n12$sw_rpp, hpc_nb_n12$sw_rpp),
HPC_Z_SW_midpoint = c(hpc_hnb_n12$sw_mpp, hpc_nb_n12$sw_mpp),
HPC_Z_AOV_lp = c(hpc_hnb_n12$aov_rpp_all[["AOV[lp]"]], hpc_nb_n12$aov_rpp_all[["AOV[lp]"]]),
HPC_Z_AOV_depth = c(hpc_hnb_n12$aov_rpp_all[["AOV[depth]"]], hpc_nb_n12$aov_rpp_all[["AOV[depth]"]]),
check.names = FALSE
)

predcheck_nonlinear_table %>%
  mutate(`Func.Form` = ifelse(dplyr::row_number() > 1, "", `Func.Form`)) %>%
  gt() %>%
  cols_label(
    `Func.Form` = "Func. Form",
    Family = "Family",
    PPC_Chisq = md("$\\chi^2$"),
    PPC_Z_SW_randomized_mean = "Z-SW RSP mean",
    PPC_Z_SW_midpoint = "Z-SW MPP",
    PPC_Z_AOV_lp = "Z-AOV-lp",
    PPC_Z_AOV_depth = "Z-AOV-depth",
    HPC_Chisq = md("$\\chi^2$"),
    HPC_Z_SW_randomized_mean = "Z-SW RSP mean",
    HPC_Z_SW_midpoint = "Z-SW MPP",
    HPC_Z_AOV_lp = "Z-AOV-lp",
    HPC_Z_AOV_depth = "Z-AOV-depth"
  ) %>%
  tab_spanner(
    label = "PPC",
    columns = c(PPC_Chisq, PPC_Z_SW_randomized_mean, PPC_Z_SW_midpoint,
      PPC_Z_AOV_lp, PPC_Z_AOV_depth)
  ) %>%
  tab_spanner(
    label = "HPC",
    columns = c(HPC_Chisq, HPC_Z_SW_randomized_mean, HPC_Z_SW_midpoint,
      HPC_Z_AOV_lp, HPC_Z_AOV_depth)
  ) %>%
  fmt_number(
    columns = c(PPC_Chisq, PPC_Z_SW_randomized_mean, PPC_Z_SW_midpoint,
      PPC_Z_AOV_lp, PPC_Z_AOV_depth,
      HPC_Chisq, HPC_Z_SW_randomized_mean, HPC_Z_SW_midpoint,

```

Table 4: PPC diagnostic p-values from full-data fits and HPC diagnostic p-values from training-data fits evaluated on the held-out data for the nonlinear HNB and NB models.

Func. Form	Family	PPC							
		χ^2	Z-SW	RSP mean	Z-SW MPP	Z-AOV-lp	Z-AOV-depth	χ^2	Z-SW R
Nonlinear	HNB	0.027		0.309	0.170	0.452	0.586	0.342	
	NB	0.171		0.157	0.394	0.379	0.376	0.842	

```

HPC_Z_AOV_lp, HPC_Z_AOV_depth),
  decimals = 3
) %>%
cols_width(`Func.Form` ~ px(95), Family ~ px(70)) %>%
cols_align(align = "left", columns = c(`Func.Form`, Family)) %>%
tab_options(table.align = "center")

```

5.7 Comparison with LOO elpd

Finally, a comparison of the models using the LOO expected log point-wise predictive density (elpd), shown in the `elpd` column of Table 5, confirms that the HNB model with a nonlinear `depth` effect significantly improves the fit over the linear version and also outperforms the NB models.

```

loo_hnb <- loo(Dolphins_hnb, moment_match = TRUE)
loo_hp <- loo(Dolphins_hp, moment_match = TRUE)
loo_nb <- loo(Dolphins_nb, moment_match = TRUE)
loo_pois <- loo(Dolphins_pois, moment_match = TRUE)
loo_hnb_nl2 <- loo(Dolphins_hnb_nl2, moment_match = TRUE)
loo_nb_nl2 <- loo(Dolphins_nb_nl2, moment_match = TRUE)
model_comparison <- loo_compare(loo_hnb, loo_hp, loo_nb, loo_pois, loo_hnb_nl2, loo_nb_nl2)

model_comparison_df <- data.frame(
  Model = c(
    "HNB (Nonlinear)",
    "HNB (Linear)",
    "NB (Nonlinear)",
    "NB (Linear)",
    "HP (Linear)",
    "Pois (Linear)"
  )
)

```

```

),
elpd_diff = c(
  model_comparison[1, 1],
  model_comparison[2, 1],
  model_comparison[3, 1],
  model_comparison[4, 1],
  model_comparison[5, 1],
  model_comparison[6, 1]
),
se_diff = c(
  model_comparison[1, 2],
  model_comparison[2, 2],
  model_comparison[3, 2],
  model_comparison[4, 2],
  model_comparison[5, 2],
  model_comparison[6, 2]
)
)

## Approximate probability of outperforming the reference model
model_comparison_df$Pr_Delta_elpd_ge_0 <- ifelse(
  model_comparison_df$se_diff > 0,
  pnorm(
    q = 0,
    mean = model_comparison_df$elpd_diff,
    sd = model_comparison_df$se_diff,
    lower.tail = FALSE
  ),
  NA_real_
)

## Formatting for display
model_comparison_df_print <- model_comparison_df %>%
  dplyr::mutate(
    elpd_diff = sprintf("%.1f", elpd_diff),
    se_diff = sprintf("%.1f", se_diff),
    Pr_Delta_elpd_ge_0 = dplyr::case_when(
      is.na(Pr_Delta_elpd_ge_0) ~ "Ref.",
      Pr_Delta_elpd_ge_0 < 0.001 ~ "<0.001",

```

```

    TRUE ~ sprintf("%.3f", Pr_Delta_elpd_ge_0)
  )
)

model_comparison_df_print %>%
  knitr::kable(
    booktabs = TRUE,
    align = c("l", "r", "r", "r"),
    col.names = c(
      "Model",
      "$\\Delta_{\\text{elpd}}$",
      "se",
      "$\\Pr(\\Delta_{\\text{elpd}} \\ge 0)$"
    ),
    escape = FALSE
  )

```

Table 5: Full Model Comparison: Expected Log Predictive Density (ELPD) Difference

Model	Δ_{elpd}	se	$\Pr(\Delta_{\text{elpd}} \geq 0)$
HNB (Nonlinear)	0.0	0.0	Ref.
HNB (Linear)	-19.9	6.8	0.002
NB (Nonlinear)	-34.0	9.9	<0.001
NB (Linear)	-42.5	9.9	<0.001
HP (Linear)	-118.6	34.4	<0.001
Pois (Linear)	-803.3	101.1	<0.001

6 Summary of Model Diagnostics and Model Comparison

```

library(dplyr)
library(gt)

## Z-residual diagnostic columns
z_linear <- data.frame(
  Model = c("HNB", "NB", "HP", "Poisson"),
  Z_SW_post = c(pv_hnb, pv_nb, pv_hp, pv_pois),

```

```

Z_AOV_depth_post = c(depth_aov_hnb, depth_aov_nb, depth_aov_hp, depth_aov_pois),
Z_Chisq_post      = c(chisq_hnb_cap, chisq_nb_cap, chisq_hp_cap, chisq_pois_cap),
Z_SW_iscv         = c(pv_hnb_iscv, pv_nb_iscv, pv_hp_iscv, pv_pois_iscv),
Z_AOV_depth_iscv  = c(depth_aov_hnb_iscv, depth_aov_nb_iscv, depth_aov_hp_iscv, depth_aov_pois_iscv),
Z_Chisq_iscv      = c(chisq_hnb_iscv_cap, chisq_nb_iscv_cap, chisq_hp_iscv_cap, chisq_pois_iscv_cap)
)

z_nonlinear <- data.frame(
  Model = c("HNB", "NB"),
  Z_SW_post      = c(pv_hnb_nl2_cap, pv_nb_nl2_cap),
  Z_AOV_depth_post = c(aov_depth_hnb_nl, aov_depth_nb_nl),
  Z_Chisq_post    = c(chisq_hnb_nl, chisq_nb_nl),
  Z_SW_iscv       = c(pv_hnb_nl2_iscv_cap, pv_nb_nl2_iscv_cap),
  Z_AOV_depth_iscv = c(aov_depth_hnb_nl_iscv, aov_depth_nb_nl_iscv),
  Z_Chisq_iscv    = c(chisq_hnb_nl_iscv, chisq_nb_nl_iscv)
)

z_summary <- bind_rows(z_linear, z_nonlinear)

## PPC/HPC columns: keep the same summary statistics as the main applied table.
predcheck_summary <- bind_rows(
  predcheck_linear_table %>% select(-`Func.Form`, -Family),
  predcheck_nonlinear_table %>% select(-`Func.Form`, -Family)
) %>%
  transmute(
    PPC_Z_SW      = PPC_Z_SW_randomized_mean,
    PPC_Z_AOV     = PPC_Z_AOV_depth,
    PPC_Y_Chisq   = PPC_Chisq,
    HPC_Z_SW      = HPC_Z_SW_randomized_mean,
    HPC_Z_AOV     = HPC_Z_AOV_depth,
    HPC_Y_Chisq   = HPC_Chisq
  )

## LOO comparison columns in the same row order as z_summary.
loo_summary <- data.frame(
  elpd_diff = c(
    model_comparison[2, 1], # HNB Linear
    model_comparison[4, 1], # NB Linear
    model_comparison[5, 1], # HP Linear
  )
)

```

```

    model_comparison[6, 1], # Poisson Linear
    model_comparison[1, 1], # HNB Nonlinear
    model_comparison[3, 1]  # NB Nonlinear
  ),
  se_diff = c(
    model_comparison[2, 2],
    model_comparison[4, 2],
    model_comparison[5, 2],
    model_comparison[6, 2],
    model_comparison[1, 2],
    model_comparison[3, 2]
  )
)

loo_summary$Pr_Delta_elpd_ge_0 <- ifelse(
  loo_summary$se_diff > 0,
  pnorm(
    q = 0,
    mean = loo_summary$elpd_diff,
    sd = loo_summary$se_diff,
    lower.tail = FALSE
  ),
  NA_real_
)

full_results_table <- bind_cols(z_summary, predcheck_summary, loo_summary)

rows_linear    <- 1:4
rows_nonlinear <- 5:6

full_results_table |>
  gt() |>
  cols_label(
    Model = "Model",
    Z_SW_post = "Z-SW",
    Z_AOV_depth_post = "Z-AOV",
    Z_Chisq_post = md("$Z\\text{-}\\chi^2$"),
    Z_SW_iscv = "Z-SW",
    Z_AOV_depth_iscv = "Z-AOV",

```

```

Z_Chisq_iscv      = md("$Z\\text{-}\\chi^2$"),
PPC_Z_SW         = "Z-SW",
PPC_Z_AOV        = "Z-AOV",
PPC_Y_Chisq      = md("$y\\text{-}\\chi^2$"),
HPC_Z_SW         = "Z-SW",
HPC_Z_AOV        = "Z-AOV",
HPC_Y_Chisq      = md("$y\\text{-}\\chi^2$"),
elpd_diff        = md("$\\Delta_{\\text{elpd}}$"),
se_diff          = "SE",
Pr_Delta_elpd_ge_0 = md("$\\Pr(\\Delta_{\\text{elpd}} \\ge 0)$")
) |>
tab_spanner(
  label = "Posterior Z-Residual Tests",
  columns = c(Z_SW_post, Z_AOV_depth_post, Z_Chisq_post)
) |>
tab_spanner(
  label = "ISCV Z-Residual Tests",
  columns = c(Z_SW_iscv, Z_AOV_depth_iscv, Z_Chisq_iscv)
) |>
tab_spanner(
  label = "PPC Tests",
  columns = c(PPC_Z_SW, PPC_Z_AOV, PPC_Y_Chisq)
) |>
tab_spanner(
  label = "HPC Tests",
  columns = c(HPC_Z_SW, HPC_Z_AOV, HPC_Y_Chisq)
) |>
tab_spanner(
  label = "LOO Comparison",
  columns = c(elpd_diff, se_diff, Pr_Delta_elpd_ge_0)
) |>
tab_row_group(label = "Linear models", rows = rows_linear) |>
tab_row_group(label = "Nonlinear models", rows = rows_nonlinear) |>
fmt_number(
  columns = c(
    Z_SW_post, Z_AOV_depth_post, Z_Chisq_post,
    Z_SW_iscv, Z_AOV_depth_iscv, Z_Chisq_iscv,
    PPC_Z_SW, PPC_Z_AOV, PPC_Y_Chisq,
    HPC_Z_SW, HPC_Z_AOV, HPC_Y_Chisq,

```


Table 6: Summary of diagnostic p-values and LOO comparisons for the dolphin-sighting data. Rows show four linear models and two nonlinear models. Posterior and ISCV Z-residual entries are upper-bound summaries of 500 replicated p-values. The Z-AOV tests are conducted against water depth. PPC and HPC columns report predictive check p-values. The last columns report LOO ($-\{elpd\}$), its SE, and an approximate probability of outperforming the reference model.

Model	Posterior Z-Residual Tests			ISCV Z-Residual Tests			PPC Tests			HPC Tests	
	Z-SW	Z-AOV	$Z-\chi^2$	Z-SW	Z-AOV	$Z-\chi^2$	Z-SW	Z-AOV	$y-\chi^2$	Z-SW	Z-AOV
Nonlinear models											
HNB	0.916	1.000	0.730	0.554	1.000	0.563	0.309	0.586	0.027	0.370	0.387
NB	0.311	0.633	0.876	0.027	0.635	0.571	0.157	0.376	0.171	0.339	0.425
Linear models											
HNB	0.801	0.013	0.812	0.449	0.015	0.675	0.301	0.033	0.016	0.271	0.403
NB	0.326	0.012	0.923	0.895	0.009	0.742	0.387	0.025	0.531	0.254	0.410
HP	0.000	0.022	0.000	0.000	0.020	0.000	0.000	0.041	0.001	0.023	0.416
Poisson	0.000	0.015	0.000	0.000	0.015	0.000	0.000	0.014	0.000	0.000	0.475

```

    elpd_diff, se_diff
  ),
  decimals = 3
) |>
fmt(
  columns = Pr_Delta_elpd_ge_0,
  fns = function(x) {
    dplyr::case_when(
      is.na(x) ~ "Ref.",
      x < 0.001 ~ "<0.001",
      TRUE ~ sprintf("%.3f", x)
    )
  }
) |>
cols_width(Model ~ px(90)) |>
cols_align(align = "left", columns = Model) |>
tab_options(table.align = "center")

```

7 Interpretation

The linear HNB and NB models fit the marginal residual distribution much better than the Poisson-based models, but their depth-based AOV diagnostics still indicate a systematic residual pattern. After the piecewise quadratic depth terms are added, the nonlinear HNB model no longer shows the same residual mean pattern against depth and also has the best LOO elpd among the candidate models. The PPC and HPC summaries provide useful comparisons, but the traditional data-scale $y\text{-}\chi^2$ check can behave differently from residual-based diagnostics in sparse count data with many zeros.

In a full applied analysis, it is also useful to check whether the conclusions are robust to prior choices and to the number of bins used in the grouped AOV diagnostic.